Smooth dogfish (*Mustelus canis*)

Summary Scores

- **Sensitivity:** Moderate
- **Exposure:** Moderate
- **Vulnerability:** Moderate

Sensitivity

Sensitivity Attributes	Score	LH DQ	Tally
Adult Mobility	1.08	3	
Complexity in Reproductive Strategy	2.36	2.5	
Dispersal of Early Life Stages	1.4	3	
Egg and Larval Survival and Settlement Requirements	1	3	
Historical Variability in Temperature	1.44	3	
Juvenile Habitat Requirements	2.56	2.75	
Other Stressors	2.68	2.8	
Population Growth Rate	2.2	2.6	
Prey Specificity	2.04	2.8	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	1.76	2.4	
Spawning Cycle	1.88	2.6	
Stock Size/Status	1.08	3	
Total Sensitivity | Certainty	**Moderate (2) | 0.69**		

Exposure

Exposure Factors	Mean	SD	Tally
Bottom Temperature	2.61	1.31	
Bottom Oxygen	2.69	1.25	
Bottom Salinity	3.23	0.64	
Bottom Aragonite Solubility	1.33	0.86	
Net primary production	1.19	0.33	
Downward Flux of Particulate Organic Carbon	1.33	0.43	
Total Exposure	**Moderate (2.37)**	**1.11**	

Vulnerability

Vulnerability	Mean	SD	Tally
Total Vulnerability	**Moderate (4.74)**	**2.22**	

Low Moderate High Very High

Figure 1: Summary Sensitivity, Exposure, and Vulnerability Scores and Score Distributions for Smooth dogfish. Exposure and Vulnerability scores are averaged spatially across the species distribution and within stocks if applicable (see Figure 5 and Figure 4). Life History Data Quality (LH DQ) and Standard Deviations (SD) are reported for senstivity and exposure/vulnerability scores, respectively.

Changes from Previous Assessment(s)

Table 1: Changes to Vulnerability, Exposure, and Sensitivity scores relative to Previous Assessments. If a citation for the previous assessment is not provided, it comes from Hare et al. (2016).

Metric	Previous Assessment(s)	Current Assessment	Change
Sensitivity	Low (Hare et al. 2016) (Loughran et al. 2025)	Moderate	Increase
Exposure	High (Hare et al. 2016) Very High (Loughran et al. 2025)	Moderate	Decrease
Vulnerability	Low (Hare et al. 2016) Moderate (Loughran et al. 2025)	Moderate	Decrease - None
Directional Effect	Neutral (Hare et al. 2016) (Loughran et al. 2025)	Neutral	None

Stocks

Smooth dogfish is managed as a single stock or stock information was not considered for this iteration of the CVA.

Score Descriptions:

Sensitivity:

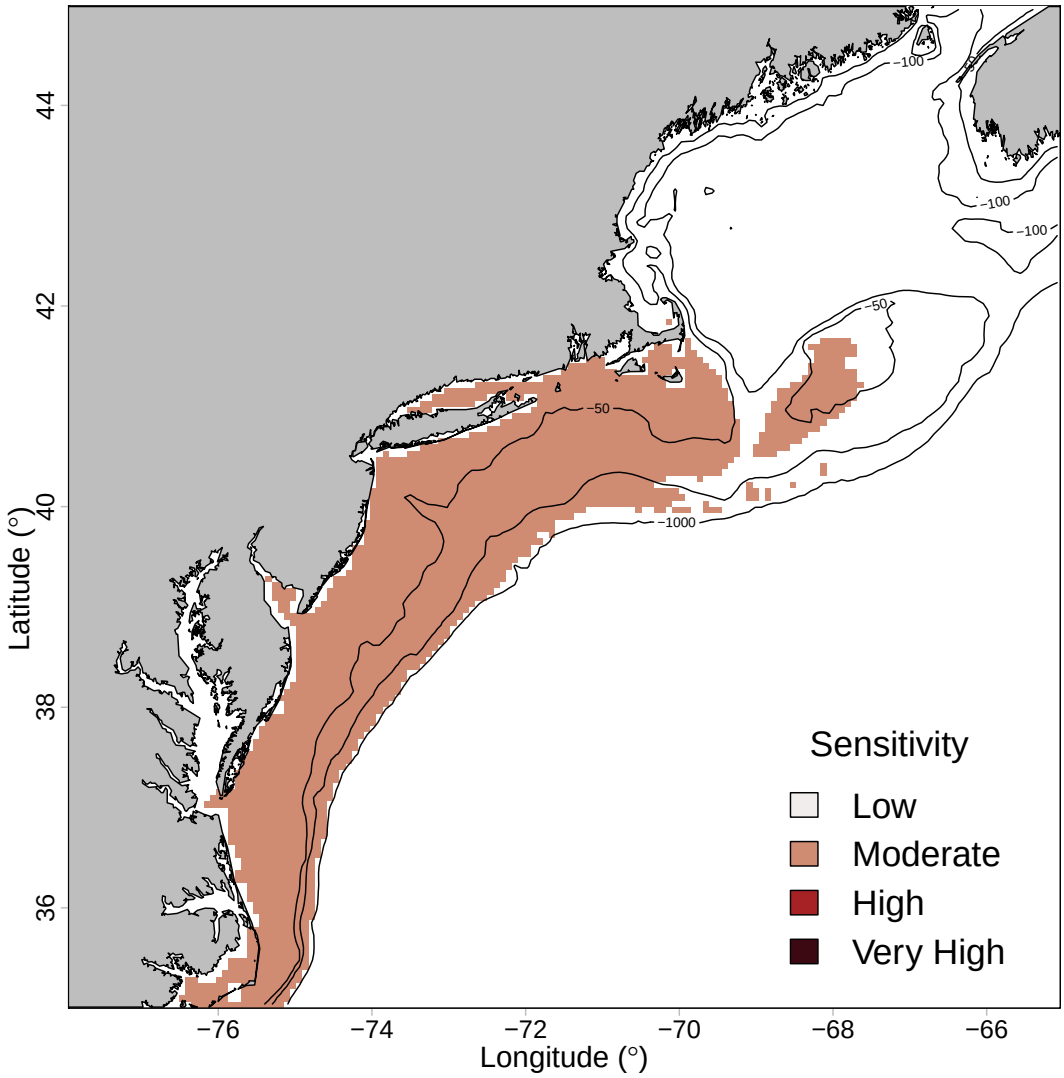


Figure 2: Smooth dogfish Sensitivity on the Northeast Shelf

Smooth dogfish received a **Moderate** sensitivity score, with a certainty of 69% according to bootstrap analysis. Two attributes had scores greater than 2.5: Other Stressors (2.7) and Juvenile Habitat Requirements (2.6) (Figure 1).

Smooth dogfish rely on coastal habitats throughout their life-cycles, which are impacted by eutrophication and hypoxia, changes in freshwater input, and increases in turbidity due to activities such as dredging (Plumlee et al. 2018). Juveniles in particular rely on nursery estuary and seagrass bed habitats, which are vulnerable habitats (Guyant et al. 2022).

In the previous Northeast Climate Vulnerability Assessment, smooth dogfish received a Low sensitivity score. This was driven by Population Growth Rate, which only received a score of 2.2 in our assessment (Hare et al. 2016). The Atlantic stock of Smooth dogfish were also assessed in the recent Highly Migratory

Species Climate Vulnerability Assessment, where it also received a Low score because no attributes scored higher than 2.5 (Loughran et al. 2025).

Exposure:

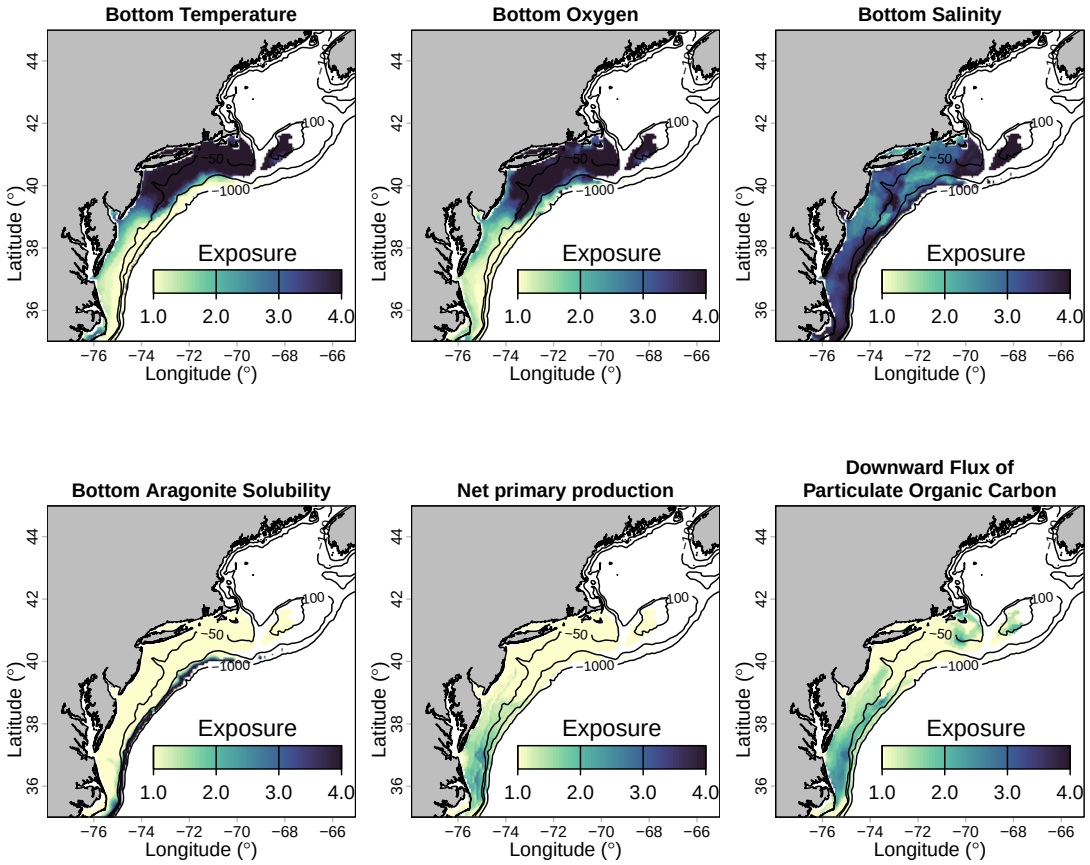


Figure 3: Maps of each Exposure Factor for Smooth dogfish. Exposure was calculated between 2014-2023 and 2025-2035

Mean Exposure score for smooth dogfish was **Moderate** (2.37 +/- 1.11) (Figure 1). Exposure was Very High over Georges Bank and south of Cape Cod. Scores were High in Southern New England and along the shelf break (Figure 4). These patterns were driven by high bottom temperature, oxygen, salinity, and aragonite saturation exposures (Figure 3).

In the previous Assessment for the Northeast Shelf, smooth dogfish received a High Exposure score, driven by sea surface temperature and ocean acidification (Hare et al. 2016). In the recent Highly Migratory Species Assessment, smooth dogfish received a Very High exposure score, driven by bottom salinity, bottom temperature, pH and sea surface oxygen (Loughran et al. 2025).

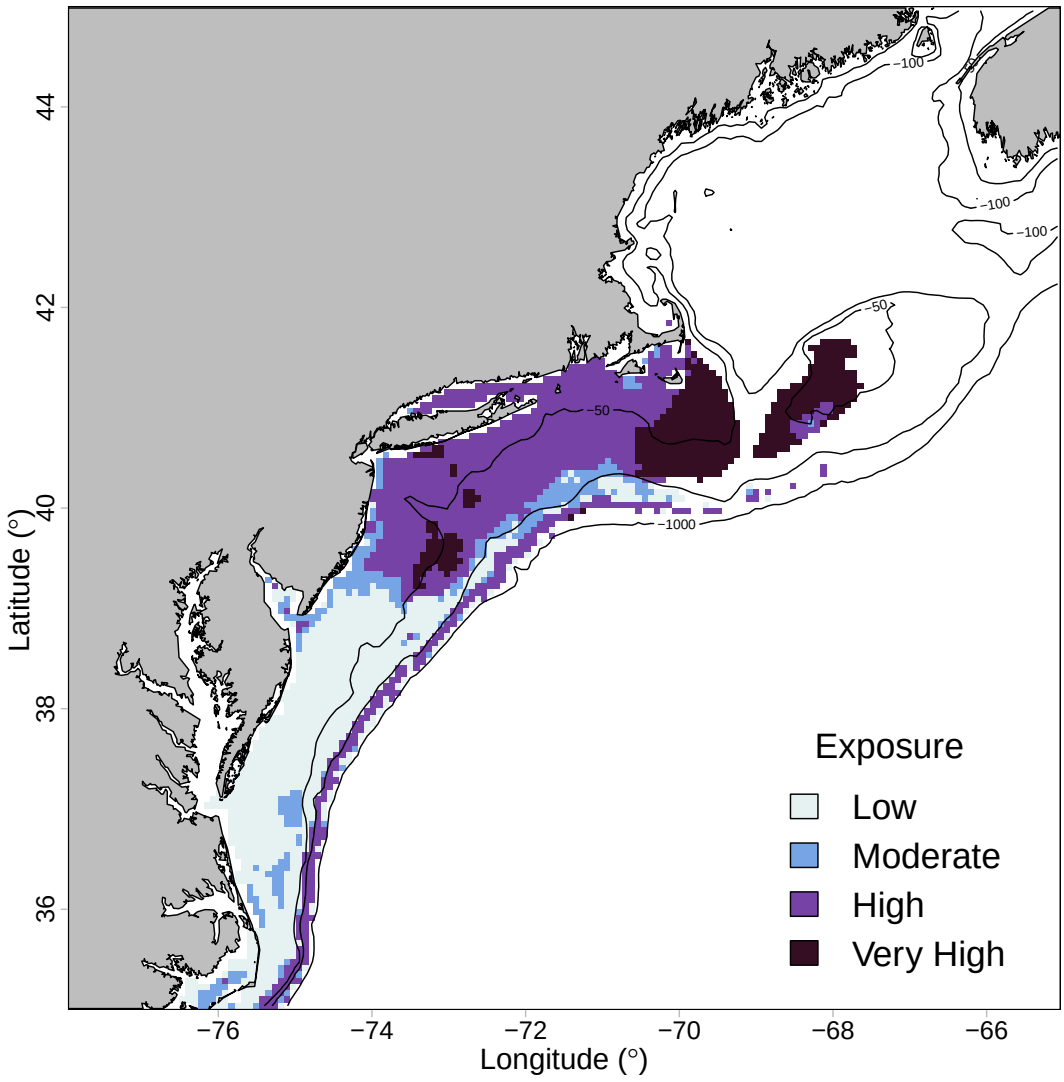


Figure 4: Total Exposure on the Northeast Shelf for Smooth dogfish. Total Exposure is calculated by applying the logic rule from Hare et al. (2016) within grid cells across exposure factors (see Figure 3)

Vulnerability:

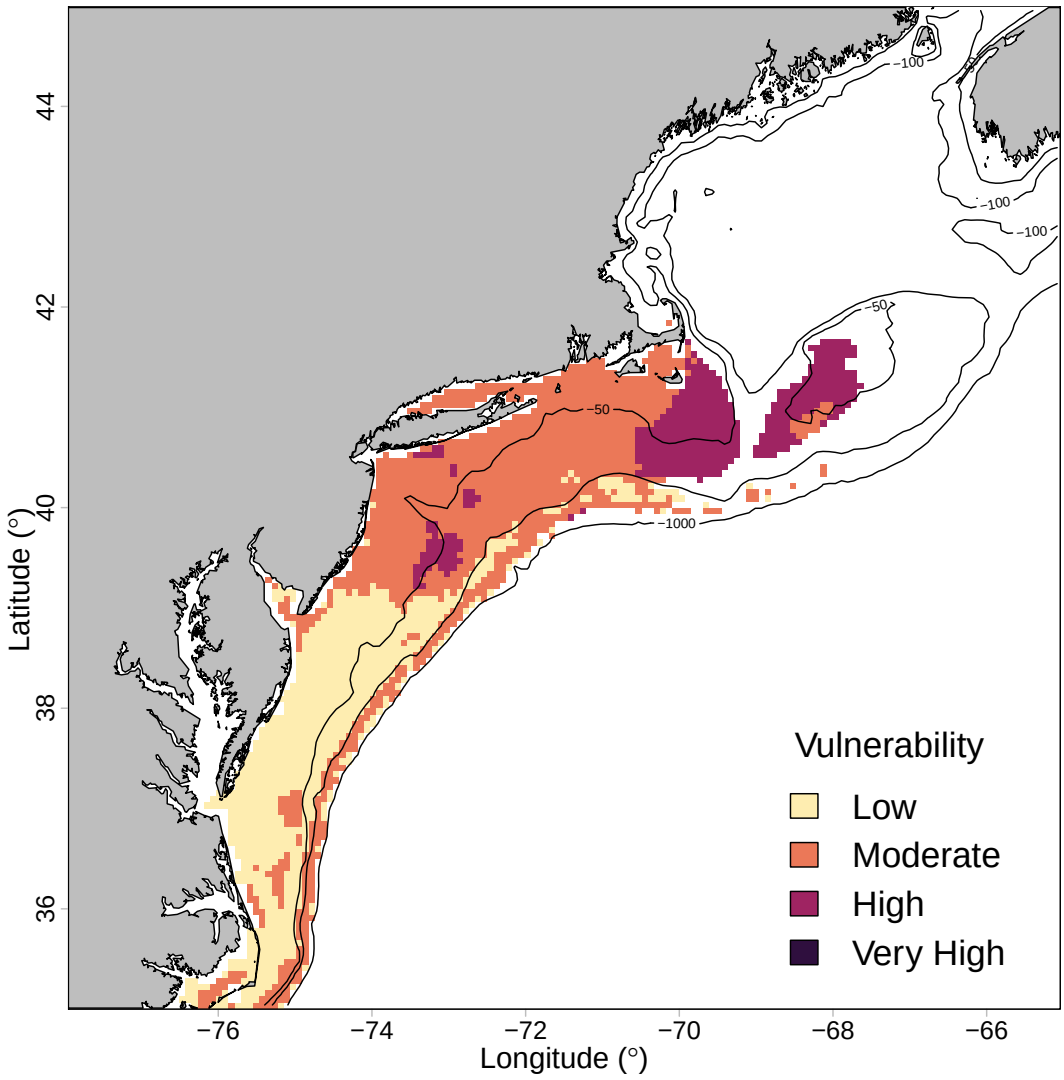


Figure 5: Vulnerability on the Northeast Shelf for Smooth dogfish. Vulnerability is the product of sensitivity and exposure (see Figure 4 and Figure 2)

Mean overall Vulnerability for smooth dogfish was Moderate (4.74 +/- 2.22) (Figure 1). Vulnerability was High south of Cape Cod and over Georges Bank and Moderate in Southern New England (Figure 5).

In the previous Assessment for the Northeast Shelf, smooth dogfish received a Low score (Hare et al. 2016). In the recent Highly Migratory Species assessment, smooth dogfish received a Moderate score, similar to our assessment (Loughran et al. 2025).

Data Quality Metrics

Life History Data Quality:

95% of data quality scores were greater than or equal to 2, indicating excellent data quality. No data gaps (mean attribute data quality score < 2) were identified (Figure 1).

Model Confidence Score:

The smooth dogfish ensemble model received a 2.2 (+/- 0.5) mean model confidence score, indicating moderate confidence in the model. Experts agreed that the model captured the distribution of smooth dogfish well. They highlighted that the model failed to capture smooth dogfish presence in the southern Gulf of Maine. The ensemble model AUC score was 0.92, indicating excellent model performance.

Additional Metrics

Distribution Shifts:

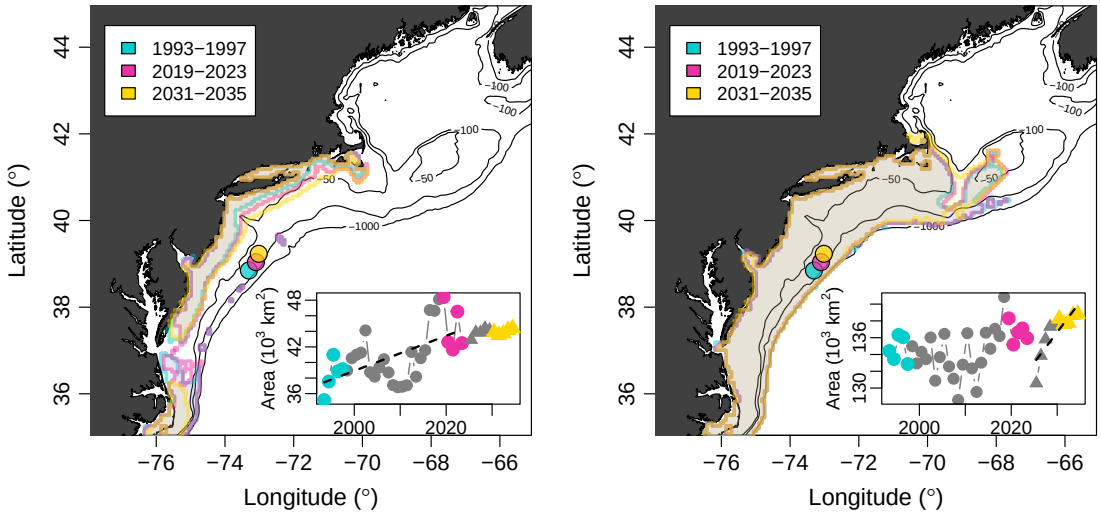


Figure 6: Distribution shifts on the Northeast Shelf for Smooth dogfish. Polygons illustrate core habitat (left) and species range (right) for five-year average periods. Points illustrate weighted centroid, calculated as the average position with species distribution models as weights. Inset timeseries illustrate the shift in core and habitat range area over time. Circles represent points from the MOM6 hindcast; triangles represent points from the MOM6 forecast. Dashed lines represent significant linear regressions

Note: For an animated version of this plot, please see the online version of this document.

Smooth dogfish core habitat significantly increased between 1993 and 2023. Core habitats expanded further offshore along the continental shelf (Figure 6). The weighted centroid shifted 28 km northeast.

Under near-future ocean conditions, smooth dogfish ranges are predicted to significantly increase. Their range is predicted to shift northward into Cape Cod Bay (Figure 6). The weighted centroid is predicted to shift another 23 km north-northeast.

In the previous Assessment for the Northeast Shelf, smooth dogfish received a Very High Distributional Vulnerability rank due to their high mobility and large seasonal migrations (Hare et al. 2016). Distributional Vulnerability Rank was not calculated in the Highly Migratory Species Assessment (Loughran et al. 2025).

Table 2: Distribution shifts based on five-year average distributions. Changes are calculated relative to the previous time period.

Time Period	Weighted Centroid Shift (km & bearing)	Core Habitat Area (change) (km^2)	Habitat Total Area (change) (km^2)	Latitude Range (change) (deg)	Change in Longitude Range (deg)
1993-1997	NA	38,866	134,934	5.99	6.65
2019-2023	27.8 km NE	44,630 (+5,764)	137,060 (+2,126)	5.86 (-0.12)	6.81 (+0.16)
2031-2035	22.5 km NNE	43,921 (-709)	138,278 (+1,218)	5.86 (0)	6.57 (-0.24)

Directional Effect:

Experts predict that climate change will have a **neutral** impact on smooth dogfish with 96% certainty. While smooth dogfish occupy estuarine habitats that are more likely to be degraded and are highly vulnerable (Guyant et al. 2022) (Plumlee et al. 2018), smooth dogfish may be tolerant of future warming and could possibly expand their range (Bockus et al. 2020). Both the previous Climate Vulnerability Assessment in the Northeast and the Highly Migratory Species Assessment also predicted that climate change would have a neutral impact on climate change (Hare et al. 2016) (Loughran et al. 2025).

Climate Effects on Abundance and Distribution

Little information exists on the potential effects of climate on smooth dogfish abundance and distribution. Coastal areas, including estuaries and smooth dogfish nursing grounds, are affected by anthropogenic activities that degrade a portion of smooth dogfish habitat. These areas are also more likely to be impacted by hypoxia and changes to freshwater input/increases in turbidity (Plumlee et al. 2018). Smooth dogfish exposed to higher temperatures (increase of 6°C) did not result in a thermal stress response, indicating a possible tolerance and ability for range expansion (Bockus et al. 2020).

Life History Synopsis

Smooth dogfish are a demersal, highly migratory shark species (HMS). The Northwest Atlantic population of smooth dogfish ranges from Cape Cod to Florida and the northern Gulf of America (Conrath, Gelsleichter, and Musick 2002). The species undergoes temperature-driven seasonal migrations north in the summer and south in the winter. They overwinter in Chesapeake Bay to South Carolina from January to May, migrating north to Southern New England beginning in early spring to give birth to pups (Castro 2010) (Dell'Apa et al. 2018). Females ovulate between June and July, with gestation lasting 10 months. Birth occurs in southern New England between early May to mid-July (Castro 2010). Once females have given birth in nearshore or estuarine areas,, young of year and immature age classes may remain in nursery grounds while females migrate north to suitable temperatures (Bangley and Rulifson 2017). Smooth dogfish prefer temperatures 9-32°C. They begin appearing in the mid-Atlantic and New England

when temperatures are greater than 6-7°C, and are most abundant along the East Coast between 13-15°C (Bockus et al. 2020). Young of year individuals use shallow shoals and marsh creek habitats particularly during night hours. Juveniles are found throughout temperate coastal estuaries, creeks, and seagrass beds, which may serve as nursery grounds (Castro 2010). Juveniles prey on small shrimp, polychaete worms, and crabs, and may fill an important predatory role during times of year when other piscivores are rare (Rountree and Able 1996). Adult smooth dogfish are bottom feeders, primarily consuming crustaceans and mainly feeling at night. They are described as a trophic specialist due to the morphological and functional modifications that allow them to feed on hard-shelled/exoskeleton-bearing prey (Castro 2010) (Bangley and Rulifson 2017). Smooth dogfish are prey items for a variety of larger sharks and fishes (Gelsleichter, Musick, and Nichols 1999). Age at maturity is 2-3 years for males and 4-7 years for females, with a maximum age of 11 years for males and 13 years for females (Conrath, Gelsleichter, and Musick 2002) (Giresi et al. 2013). In state waters, smooth dogfish are managed by the Atlantic States Marine Fisheries Commission's Interstate Fishery Management Plan for Coastal Sharks (Atlantic States Marine Fisheries Commission 2024). NOAA Fisheries produces an annual Stock Assessment and Fisheries Evaluation (SAFE) Report detailing the status of HMS in federal waters (National Marine Fisheries Service 2023). The current stock status is not overfished and overfishing is not occurring (Southeast Data, Assessment, and Review 2015).

Supplemental Scores

Exposure Timeseries

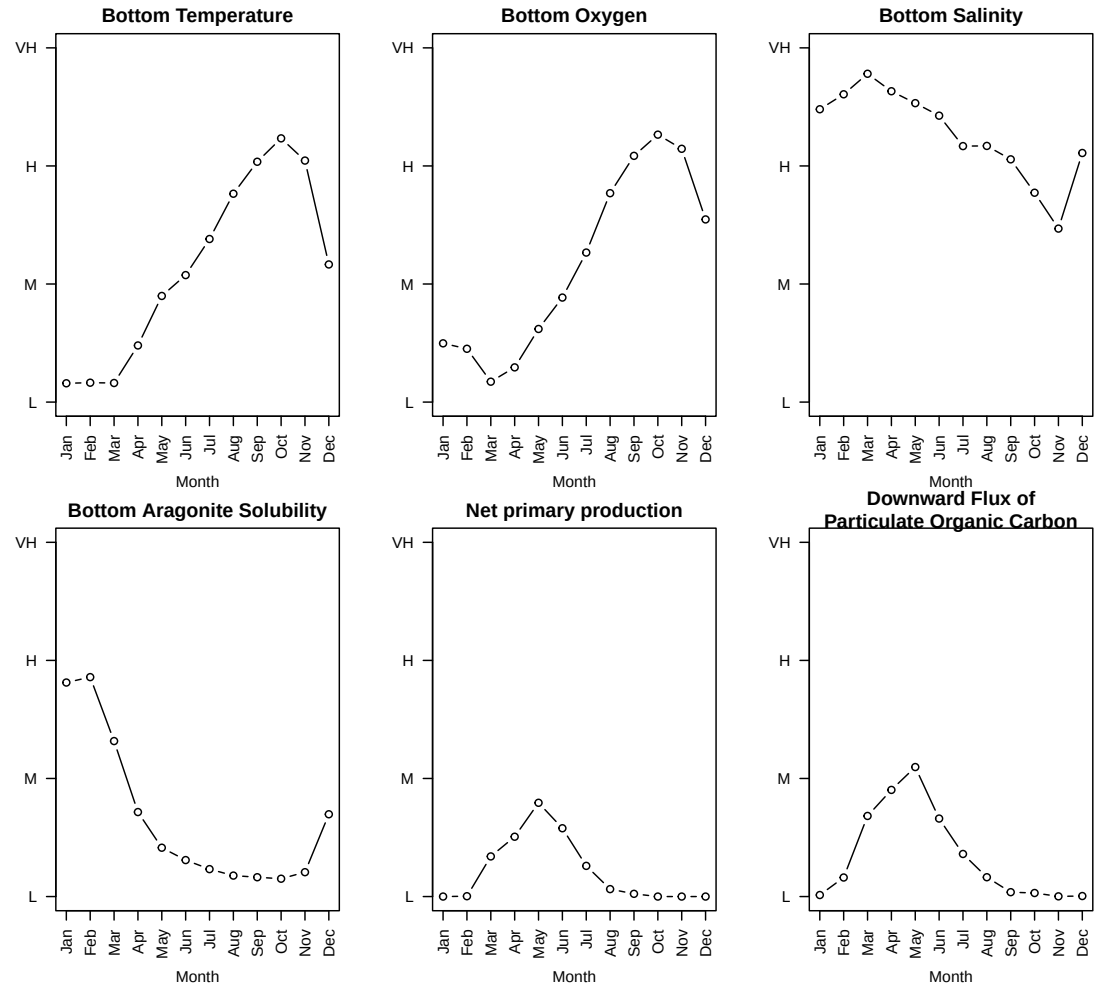


Figure 7: Monthly Timeseries of each Exposure Factor for Smooth dogfish. Exposure was calculated between 2014-2023 and 2025-2035

Across months, exposure ranged from Low to High. Exposure was lowest from March through July, and greatest from September through November (Figure 8). This was driven by changes in bottom temperature and oxygen exposure, which also peaked September through November (Figure 7).

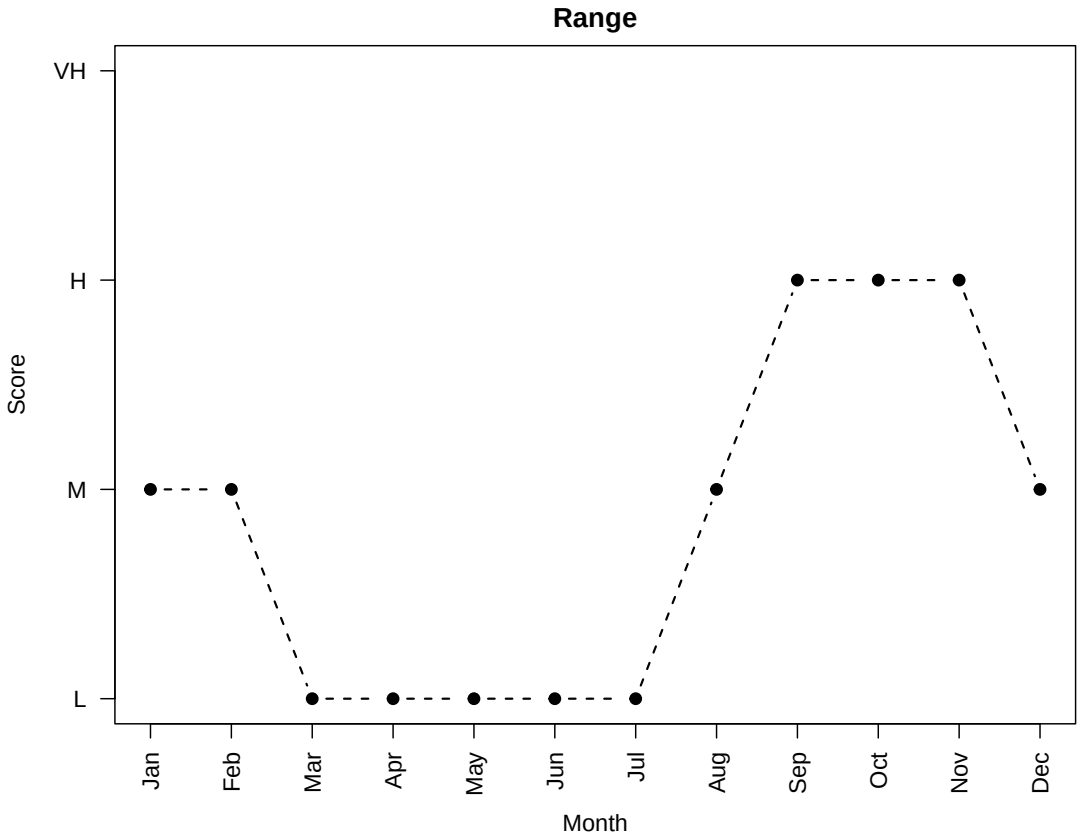


Figure 8: Timeseries of Total Exposure for Smooth dogfish

Exposure due to Significant Factors

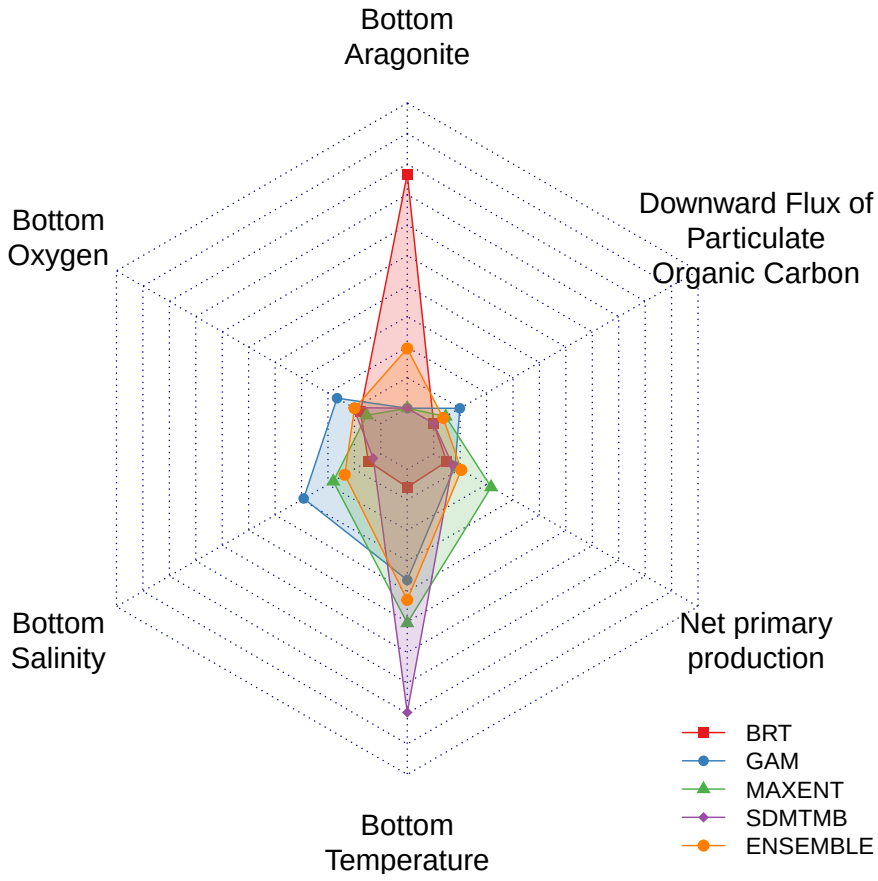


Figure 9: Dynamic Factors driving Smooth dogfish distributions. The closer the polygon is to the edge of the figure, the more important the variable is. Each ring represents 10% variance explained. Factors must explain at least 10% of variance in the ensemble model to be used to calculate total exposure.

Bottom temperature, salinity, and aragonite solubility, as well as net primary production were the major dynamic drivers of smooth dogfish distributions (Figure 9). Smooth dogfish are demersal and their migrations are cued by temperature (Castro 2010) (Dell'Apa et al. 2018). Juveniles use coastal estuaries and other nearshore habitats as nursery habitat (Castro 2010). Crustaceans are the predominate prey item for adults (Castro 2010) (Bangley and Rulifson 2017). When these factors were used to calculate exposure, exposure scores remained Moderate (1.91 +/- 0.90) (Figure 11). Scores were still High over Georges Bank, south of Cape Cod and Long Island, and on the continental shelf break (Figure 10).

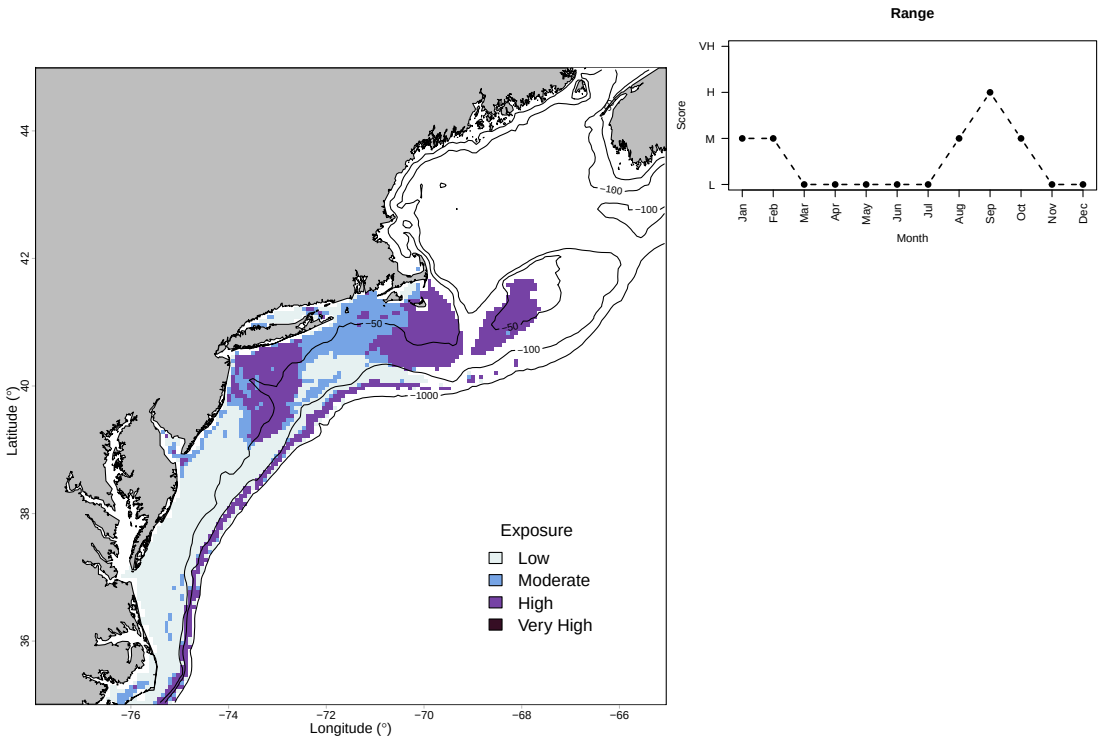


Figure 10: Total Exposure Map and Timeseries when Exposure was calculated with only Exposure Factors critical to Smooth dogfish distributions

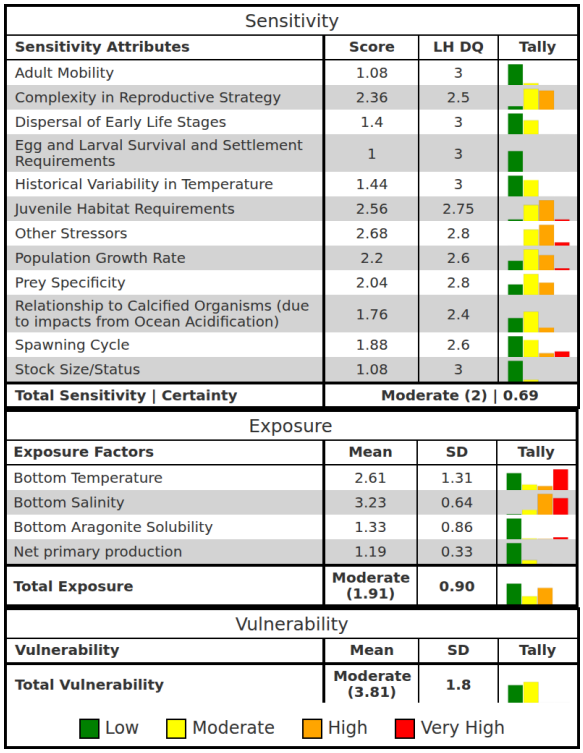


Figure 11: As in Figure 1 for Smooth dogfish, but with only significant Exposure factors and the resulting Vulnerability Scores.

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